

SIMULATION COMMERCIAL BANK | QUANTITATIVE RISK MANAGEMENT

FORMAL MODEL VALIDATION REPORT

Commercial Real Estate (CRE) Probability of Default (PD) Model

Compliance Framework: Federal Reserve SR 11-7 (Model Risk Management)

Model Risk Tier: Medium

Date of Validation: July 2026

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Document Control & Version History

Version	Date	Author	Description of Changes
1.0	July 2026	Nicolas Lucero	Initial Validation under SR 11-7 standards.

MODEL VALIDATION REPORT: CRE Probability of Default (PD)

Date of Review: July 2026

Model Owner: Quantitative Risk Management

Validator: Nicolas Lucero, Independent Quantitative Analyst

1. Executive Summary

- **Model Name:** Commercial Real Estate (CRE) 12-Month Probability of Default (PD) Model
 - **Validation Conclusion:** Approved with Conditions
 - **Overall Risk Rating:** Medium

Validation Abstract

An independent validation was conducted on the CRE PD Model in accordance with Federal Reserve SR 11-7 guidelines. The model demonstrates acceptable in-sample fit and out-of-sample rank-ordering capability for baseline credit monitoring. However, the validation identified significant non-linear tail risk under macroeconomic stress. While the baseline probability of default increases linearly with unemployment, simultaneous shocks to asset valuations (LTV) cause a compounding deterioration in Loss Given Default (LGD), resulting in a 5x increase in Expected Loss (EL).

Key Findings & Conditions for Approval

1. **Condition 1 (Boundary Guardrails):** The model violates the linearity of log-odds assumption when Loan-to-Value (LTV) exceeds 90%. Manual underwriting overrides must be triggered for any origination exceeding this threshold.
2. **Condition 2 (Data Integrity):** The validation uncovered pipeline vulnerabilities accepting negative Debt Service Coverage Ratios (DSCR). Hard-coded assertions must be implemented in the ETL pipeline before the next production run.

2. Model Purpose & Use Scope

The model is a statistical classification tool designed to estimate the 12-month forward-looking Probability of Default (PD) for a portfolio of commercial real estate loans.

- **Approved Use:** To calculate baseline capital requirements, rank-order portfolio risk, and conduct CCAR/DFAST macroeconomic stress testing.
- **Prohibited Use:** This model is strictly prohibited from being used for pricing individual retail mortgages or consumer credit products.

3. Conceptual Soundness

The underlying mathematical framework is a multivariate binomial logistic regression. It maps continuous and categorical borrower variables into a probability space between 0 and 1.

3.1 Mathematical Formulation

The log-odds of default are defined by the following equation:

$$PD = \frac{1}{1 + e^{-(\beta_0 + \beta_1 LTV + \beta_2 DSCR + \beta_3 Macro)}}$$

Validator's Assessment: The choice of logistic regression is conceptually sound for binary credit classification. It correctly bounds outputs, preventing theoretically impossible negative default probabilities, and offers high coefficient interpretability required for fair lending audits.

3.2 Variable Selection & Theoretical Limitations

Variable	Financial Rationale	Validator Assessment
LTV	Captures collateral coverage.	Sound. Represents the primary driver of strategic default.
DSCR	Captures operational cash flow.	Sound. Accurately measures short-term debt servicing ability.
Macro	Regional unemployment rate.	Sound. Links systemic economic health to borrower solvency.

Identified Limitation: The model excludes "Property Type" (e.g., Office vs. Multi-family) as a categorical variable. Given the structural bifurcation in commercial real estate post-2020, treating all asset classes with identical LTVs as sharing identical risk profiles is a conceptual weakness that may under-predict defaults in the commercial office sector.

4. Data Integrity & System Implementation

A rigorous audit of the raw data ingestion pipeline was conducted to evaluate the handling of missing, erroneous, or boundary-violating inputs.

- Missing Values:** The dataset contained dropped LTV records. The model development team successfully implemented a median-imputation protocol; however, the validation team recommends rejecting incomplete records entirely to prevent synthetic data from artificially lowering variance.
- Boundary Logic Violations:** The validation script identified records containing impossible states, including $DSCR < 0$ and unflagged $LTV > 1.05$ (indicating unsecured exposure without proper coding).
- Remediation:** The validation team applied stringent cleaning protocols, purging records outside the physically possible boundaries of the financial system prior to out-of-sample testing.

5. Outcomes Analysis & Out-of-Sample Backtesting

The model was trained on an 80% sample partition and evaluated on a pristine 20% out-of-sample holdout to test predictive reliability and prevent overfitting.

5.1 Performance Metrics

Metric	Result	Industry Standard	Status
ROC-AUC	0.762	> 0.70	Pass
Brier Score	0.084	Closer to 0 is better	Pass
P-Value (All Vars)	< 0.01	< 0.05	Pass

Validator's Assessment: The ROC-AUC score of 0.762 confirms the model possesses strong rank-ordering capability on unseen data. The coefficients behave in the expected economic directions (higher LTV increases PD; higher DSCR decreases PD).

6. Stress Testing & Expected Loss (EL)

To evaluate capital adequacy under extreme adverse conditions, the validation team subjected the out-of-sample portfolio to a severe macroeconomic shock scenario.

The Shock Scenario:

- 1. Regional unemployment is forced to a flat 12% across all exposures.
- 2. Commercial property values experience a sudden 20% depreciation, mathematically increasing the LTV of all active loans by a factor of 1.25.

6.1 Dollar-Value Financial Impact

Expected Loss is calculated using the standard regulatory framework: $EL = PD \times LGD \times EAD$. Under the stress scenario, the Loss Given Default (LGD) baseline assumption was elevated from 45% to 65% to reflect depressed recovery values during a market crash.

Metric	Baseline Scenario	Severe Adverse Scenario	Change
Average Portfolio PD	4.2%	15.8%	+ 3.7x
Watchlist (Loans > 20% PD)	42	418	+ 10.0x
Total Expected Loss (EL)	\$1.85 Million	\$9.62 Million	+ 5.2x

Validator's Conclusion:

The model successfully computes stress scenarios without mathematical failure or returning boundary-violating probabilities. However, the risk committee must note the non-linear consequence of systemic shocks: while the probability of default scales by a factor of 3.7, the simultaneous collapse in collateral values drives the actual Expected Loss (capital shortfall) up by a factor of 5.2. The bank must ensure adequate Tier 1 capital reserves to absorb this \$9.62M shocked loss.

7. Ongoing Monitoring & Governance

In compliance with SR 11-7, this model must be subject to continuous monitoring to detect performance decay or regime shifts.

- **Validation Frequency:** Annually.
- **Performance Triggers:** If the out-of-sample ROC-AUC drops below 0.70 for two consecutive quarters, the model must be quarantined and recalibrated to reflect the new macroeconomic environment.